Genome QC · genomes to atlas

Inputs: genome FASTAs + reference databases

Input · Compute · Data · Output

Preprocessing 1

Genome FASTAs → Prepare inputs (Genome QC) → Assembly statistics (SeqKit) → Contig variants

Quality & taxonomy 2

Contig variants → Completeness / quality (CheckM, Qscore) → Taxonomic assignment (GTDB-Tk) → Scored genomes

rRNA consistency 3

Classified variants → Predict rRNAs (barrnap) → Compare rRNAs (BLAST+, Genome QC) → rRNA evidence

Genome assessment 4

Assessed variants → Choose representative (Qscore, N50, length) → tRNAs / chimerism (tRNAscan-SE, GUNC) → Combined evidence

Final integration 5

All representatives → Join measurements (Genome QC) → Apply selection (User criteria) → Master + selected

Atlas & explorer 6

Master QC table → Build genome atlas (Python plotting) → Explore / select / export (Genome QC Explorer) → Figures + subsets